

ALLWIN SUDHAKARAN



Physics - M.Sc., Ph.D., Post Doc

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ABOUT ME

I am a dedicated physicist specializing in material science with strong expertise in experimental and computational research. My work bridges physics, chemistry, and materials engineering, focusing on the design and development of advanced functional materials for energy and environmental applications. Alongside my research, I am equally passionate about academic mentorship and collaborative teaching, fostering inclusive and engaging learning environments. My long-term goal is to advance impactful research while contributing to the training of the next generation of scientists through a balance of innovation, education, and collaboration.

EDUCATION

/Year of Completion

➤ Research Scientific Staff

2025- Present

Electro-Catalysis Laboratory, School of Chemistry, Indian Institute of Science Education and Research Thiruvananthapuram (IISER TVM), Thiruvananthapuram-695551. Kerala, India.

Specialization: Oxygen Evolution Reaction (OER), Oxygen Reduction Reaction (ORR), and Hydrogen Evolution Reaction (HER) in metals/metal oxides, perovskite materials, and carbon-based catalysts.

Area of Exposure: In my current role, I have gained extensive exposure to both experimental and computational aspects of electrocatalysis. My work focuses on designing and evaluating catalysts for OER, ORR, and HER, particularly in metal oxides, metals, perovskites, and carbon-based materials. I bring hands-on expertise in catalyst synthesis, electrochemical testing, and advanced material characterization techniques, while also integrating computational modeling to deepen mechanistic insights. I am proficient in density functional theory (DFT) simulations using **Gaussian, Quantum ESPRESSO, Advanced Nano-labo, and Materials Studio software**, which allows me to correlate electronic structure with catalytic performance. This combined physics-chemistry approach enables me to bridge experimental observations with theoretical analysis, providing a comprehensive understanding of catalytic processes.

➤ **Post Doctoral Fellow**

2023- 2025

Department of Nanosciene, Center for Nano and Material Sciences, Jain (Deemed-to-be) University, Jain Global Campus, Ramanagaram-562112, Karnataka, India.

Specialization: Electrocatalysis, CO₂ Utilization, Reduction, and fixation, Hydrogen Evaluation Reaction (HER), Oxygen Evolution Reaction (OER), CO₂ conversion into value-added products.

Area of Exposure: In my professional journey, I have garnered extensive exposure to a variety of sophisticated scientific instruments and techniques. My hands-on experience includes operating the CH instrument, rotary evaporator, and advanced characterization tools such as Zeta potential and Dynamic Light Scattering (DLS). I have also honed my skills in the electrospinning process, which is crucial for fabricating nanofibers.

My research, which specializes in electrocatalysis, focuses on the utilization, reduction, and fixation of CO₂. I am proficient in the Hydrogen Evolution Reaction (HER) and Oxygen Evolution Reaction (OER) processes, key components in the production of sustainable energy. Additionally, I have dedicated significant effort to converting CO₂ into value-added products, aiming to mitigate environmental impact while contributing to the development of green technologies.

Instruments Handled: CH instrument, rotary evaporator, Zeta potential, Dynamic light scattering (DLS), Electrospinning.

➤ **Ph.D., Physics (Material Science)**

2023

Karpagam University, Coimbatore-641021, Tamil Nadu, India.

Dissertation Title: Synthesis and Comparative Study of Aluminium and Copper Doped Barium Hexaferrite/ Cobalt Zinc Ferrite Nanocomposite Materials for Memory Device and Filter Applications

Specialization: Physics (Material Science).

Research Methodology: Sol-Gel Auto combustion, Sol-gel Citrate method, Physical Mixing method, Ball milling Method.

Analysis Techniques Employed: XRD: X-ray Diffraction, FTIR: Fourier Transform Infrared Spectroscopy, SEM: Scanning Electron Microscopy, EDAX: Energy Dispersive X-ray Spectroscopy, UV: Ultraviolet, PL: Photoluminescence, Raman: Raman Spectroscopy, Dielectric Studies, VSM: Vibrating Sample Magnetometer.

Areas of Exposure: Material Science namely, metals and composites, powders, Nanocomposites, Ferrites, Oxide materials, Magnetic materials, Thin Films, sintering, processing, mechanical alloying, phases and microstructures characterization, Physical properties testing and characterization, heat treatments etc.

Thesis Overview: My research advanced material science, focusing on applications in memory devices and filter technologies. I observed a significant correlation between dopant concentration, microstructure, and magnetic performance. Precise control over dopant levels was found to influence magnetic properties, suggesting tailored applications in nanoscale magnetic devices. This study provides critical insights into designing, synthesizing, and characterizing advanced nanocomposite materials for next- generation memory devices and high-performance filters, enhancing our understanding of nanoscale magnetic systems and their practical applications in material science.

Internal Ph.D. Supervisor: Dr. E. Sivasenthil, Associate Professor, Advance Functional and Energy Materials Laboratory, Department of Physics, Karpagam University, Coimbatore, 641 021, Tamil Nadu, India.

External Expert Member: Dr. S. Arumugam, Vice Chancellor, Tamil Nadu Open University, Chennai - 600015, Tamil Nadu, India.

➤ **M.Sc., Physics: 95.4 % (First Class with Distinction)**

2019

Karpagam University, Coimbatore-641021, Tamil Nadu, India.

Dissertation Title: Quantitatively probing magnetic nanostructures ZnFe_2O_4 with superior spatial resolution.

Project Institution: Indira Gandhi Center for Atomic Research (IGCAR), Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India.

Specialization: Physics (Material Science), Nanomaterials

Thesis Overview:

In my MSc thesis, I focused on the quantitative analysis of magnetic nanostructures, specifically ZnFe_2O_4 , emphasizing achieving superior spatial resolution. The key findings and conclusions of my research are as follows:

- I successfully synthesized phase-pure spinel nanostructured ZnFe_2O_4 using a co-precipitation method. This synthesis method ensured the desired phase and crystalline purity of the material.
- Comprehensive characterization techniques, including Raman spectroscopy and X-ray diffraction (XRD), confirmed the phase and crystalline purity of the ZnFe_2O_4 nanostructures.
- Microstructural analysis through scanning electron microscopy (SEM) and atomic force microscopy (AFM) revealed the smallest grain size, approximately 20 nm, which is crucial for understanding the material's properties at the nanoscale.
- Magnetic Force Microscopy (MFM) imaging unveiled the presence of very tiny magnetic domains, approximately 30 nm in size, within the ZnFe_2O_4 nanostructures. These images also showed variations in brightness, indicating differences in shape, size, and orientation of the nanostructures.
- A noteworthy observation was the relationship between magnetic domain intensity and grain size. As grain size increased, magnetic domain intensity increased and then decreased beyond a certain threshold.
- For grain sizes below 20 nm, MFM imaging exhibited significant diffusion of the phase signal, providing direct evidence of super-paramagnetism in the nanostructured ZnFe_2O_4 .
- The results of this research demonstrate the significant impact of particle size on the magnetic properties of magnetic materials. This finding has promising implications for developing novel magnetic materials for nanomagnetic devices. Importantly, this work opens up the possibility of directly and quantitatively probing the weak magnetic stray fields emanating from nanoscale magnetic systems with unparalleled spatial resolution.

Areas of exposure: Nuclear physics, Nanomaterial characterization and sensors, Surface Nanoscale examination, Material science.

➤ **B.Sc., Physics: 87.4 % - (First Class with Distinction)** ----- **2017**

Karpagam University, Coimbatore-641021, Tamil Nadu, India.

Dissertation Title: Theoretical Study on Gravitational Waves and LIGO (Laser Interferometer Gravitational-Wave Observatory) as an Observatory

➤ **Higher Secondary (XII standard): 86.33 %** ----- **2014**

Hindusthan Matric Higher Secondary School, Coimbatore, Tamil Nadu, India

Subjects: Computer Science, Physics, Chemistry, Mathematics, Tamil, and English.

Board of Studies: State Board of Tamil Nadu

➤ **Secondary School (X standard): 89 %** ----- **2012**

Adwaith G.N.S Matric Higher Secondary School, Coimbatore, Tamil Nadu, India.

Subjects: Physics, Chemistry, Mathematics, Biology, Tamil, and English.

Board of Studies: State Board of Tamil Nadu.

HEADSHIP & RESPONSIBILITIES		/Duration
➤ Reviewer- Journal of Inorganic and Organometallic Polymers and Materials (IF: 3.9), <i>Springer Nature</i>	p-ISSN: 1574-1443, e-ISSN: 1574-1451	2024-Present
➤ Reviewer- Scientific Reports (IF: 3.9), <i>Springer Nature</i>	e-ISSN: 2045-2322	2024-Present
➤ Reviewer- Heliyon (IF:3.4), <i>Scopus</i>	ISSN: 2405-8440	2024-Present
➤ Reviewer- Optical and Quantum Electronics (IF: 3.3), <i>Springer Nature</i>	p-ISSN: 0306-8919, e-ISSN: 1572-817X	2024-Present
➤ Reviewer- Polymer Bulletin (IF:3.1), <i>Springer Nature</i>	p-ISSN: 0170-0839, e-ISSN: 1436-2449	2024-Present
➤ Reviewer - Al-Mustansiriyah Journal of Science	p-ISSN: 1814-635X, e-ISSN: 2521-3520	2024-Present

INTERNSHIP & TRAINING

/Duration

➤ Mendeley Training Course

14/06/2020

Siva Institute of Pharmacy, Bilaspur, India.

Job position/Title: Trainee

Description: Mastering reference management and research workflow. Topics included software introduction, citation organization, and collaboration strategies. Acquired practical skills in document annotation and seamless citation insertion. Enhanced productivity and research efficiency through effective Mendeley utilization.

➤ M.Sc Final Year project Trainee

SEP 2018 to APR 2019

Indira Gandhi Center for Atomic Research (IGCAR), Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India

Job position/Title: Project/ Research Intern

Description: Conducted research on ZnFe_2O_4 as part of M.Sc., project in collaboration with Karpagam University and IGCAR, TN, India and developed training through hands-on experimentation and data analysis via Scanning Probe Microscope (SPM) - (Atomic & Magnetic Force Microscopy (AFM & MFM))

➤ Summer Training Internship for Physics and Chemistry (STIPAC)

APR 2018 to JULY 2019

Indira Gandhi Center for Atomic Research (IGCAR), Department of Atomic Energy (DAE), Kalpakkam, Tamil Nadu, India

Job position/Title: Research Intern

Description: Conducted research entitled, “Nanoscale study of nanocomposite of $\text{BaTiO}_3 \times \text{ZnFe}_2\text{O}_4$ ” and exposed areas including Nuclear Physics, Nuclear power plant, Material Science, Nanomaterials, quantum optics, MEMS & NEMS.

FIELD OF INTEREST

- | | | |
|-----------------------------|-------------------|----------------------|
| ✓ Material Science | ✓ Metal Oxides | ✓ Atomic & Molecular |
| ✓ Nanocomposites | ✓ Charge Density | ✓ Condensed matter |
| ✓ Multifunctional materials | ✓ Microstructures | ✓ Thin films |
| ✓ Ferrites | ✓ Solid State | ✓ Relativity |
| ✓ Green synthesis | ✓ Nanoscience | ✓ Electrocatalysis |

JOURNAL PUBLICATIONS

S.No.	Title	Journal Name/ (National/ International)	Impact factor/ Indexing	Status/ Vol., Year, Page No.	Document link
1	Advances in Electrochemical CO ₂ Reduction to CO for Enabling A Sustainable Energy Economy	Progress in Materials Science/ International	40.8/SCI, Scopus & Web of Science	Published/ May 2026: 101655	View article
2	Advancements and Perspective of Environmentally Sustainable Technologies for Electrochemical Selective Conversion of CO ₂ to Methanol	Catalysis Reviews/ International	13.6/ SCI, Scopus & Web of Science	Published/ May 2024: 1-113	View article
3	Topical advancement in electrochemical conversion of carbon dioxide to methane: recent advances, challenges, and prospects for carbon neutrality	Fuel (Elsevier)/ International	7.8/ SCI, Scopus & Web of Science	Published/ Vol 430, Part E, March 2027: 141466	View article
4	Unraveling the in-situ engineered mixed metal oxides anchored on activated carbon cloth as a flexible electrode material for electrochemical reduction of CO ₂ into formic acid	Sustainable Materials and Technologies/ International	8.7/SCI, Scopus & Web of Science	Published/ Vol 43, April 2025: e01260	View article
5	Unlocking the potential of in-situ thermally reduced graphene oxide anchored with ternary metal oxide-spinel catalyst for sustainable high-performance electrochemical valorisation of CO ₂	Fuel (Elsevier)/ International	7.8/ SCI, Scopus & Web of Science	Published/ Vol 387, May 2025: 134458	View article
6	Unleashing the potential of N-doped metal-free bi-functional hollow carbon spheres: Catalyst for environmental conservation and advancing sustainable energy solutions	Fuel (Elsevier)/ International	7.8/ SCI, Scopus & Web of Science	Published/ Vol 374, Oct 2024: 132457	View article

7	Recent Advances in the Fixation of CO ₂ into Quinazoline and Benzimidazole	Energy and Fuels (ACS)/ International	5.3/ SCI, Scopus & Web of Science	Published/ Vol 38 (10), April 2024: 8481-8515	View article
8	Structure-property connections in dual-phase multifunctional barium titanate-based nanocomposites: Insights from charge density distributions	Materials Chemistry and Physics/ International	4.6/ SCI, Scopus & Web of Science	Published/ Vol 318, May 2024: 129290	View article
9	Multifunctionality of AlBaFe ₁₂ O ₁₉ /CoZnFe ₂ O ₄ hybrid nanocomposite: Promising structural, elastic, morphological, compositional, optical, and magnetic properties	Journal of Physics and Chemistry of Solids/ International	4.3/ SCI, Scopus & Web of Science	Published/ Vol 174, Mar 2023: 111134	View article
10	In-Situ MgO-engineered dendritic fibrous nano silica as an efficient catalyst for fixation of CO ₂ at atmospheric pressure	Chemical Engineering Science/ International	4.3/ SCI, Scopus & Web of Science	Published/ Vol 317, Nov 2025:122068	View article
11	Optical Bandgap characterization of Novel Nano scale CBM-CZF composite	Bulletin of Environment, Pharmacology and Life Sciences (BEPLS)/ International	4.0/ Web of Science & UGC	Published/ Special Issue [1], May 2022: 643-650	View article
12	Deep Eutectic Solvent Engineered Dendritic Fibrous Nano-Silica Catalyst for Sustainable Fixation of CO ₂ Into Value-Added Product at Atmospheric Condition	Applied Organometallic Chemistry/ International	3.7/ SCI, Scopus & Web of Science	Published/ Vol 39, Jan 2025:e7910	View article
13	Investigating structure, elasticity, morphology, composition, and optical behavior of Al-doped BaFe ₁₂ O ₁₉ /CoZnFe ₂ O ₄ hybrid composite	Journal of Materials Research/ International	2.9/ SCI, Scopus & Web of Science	Published/ Vol 38, Jan 2023:1239 –1253	View article

14	Electron density effect of hybrid ferrites on structural, spectroscopic, optical, and magnetic properties in $(\text{BaTiO}_3)_{(1-x)} + (\text{ZnFe}_2\text{O}_4)_x$ nanoceramics	Journal of Material Science: Materials in Electronics/ International	2.8/ SCI, Scopus, & Web of Science	Published/ / Vol 34, Jan 2023: 104	View article
15	Potential biological and optoelectronic applications of AgO:ZnO nanocomposite synthesized by green approach	The European Physical Journal Plus/ International	2.8/SCI, Scopus, & Web of Science	Published/ / Vol 139, Dec 2024: 118	View article
16	Investigating Optical Properties of Novel Heterostructure Aluminium Doped Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite.	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ / Vol 13 Issue 71, Apr 2022: 39368-39375	View article
17	A Topological Control Algorithm for Decision-Making in Complex Material Systems	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ / Vol 15 Issue 85, Aug 2024: 1-11	View article
18	Synthesize of ZnO and CuO nanoparticles with plasma jet at different treatment times and testing its optical parameters with UV-Vis-NIR	Applied Physics A: Materials Science & Processing (Springer)/ International	2.6/ SCI, Scopus, & Web of Science	Published/ / Vol 130, Jul 2024: 533	View article
19	Biological activity of MgO nanoparticle synthesis by plasma-assisted reduction method	Physica Scripta (IOP Science)/ International	2.6/ SCI, Scopus, & Web of Science	Published/ / Vol 99, Number 11 Oct 2024: 115901	View article
20	An Efficient and Selective Synthesis of Chalcones via Claisen-Schmidt Condensation Reaction Catalyzed by CuO@rGO Nano Catalyst	Catalysis Letters/ International	2.4/ SCI, Scopus, & Web of Science	Published/Vol 155, Sep 2025: 327	View article

21	Comprehensive Analysis of Optical, Dielectric and Electrical properties in Al- and Cu- Doped Barium Hexaferrite/Cobalt-Zinc Ferrite Hybrid Nanocomposites	Journal of Electronic Materials/ International	2.1/ SCI, Scopus, & Web of Science	Published/ / Vol 53 Issue 06, Jun 2024: 39368-39375	View article
22	Deliberating Structural, Optical and Spectroscopic Properties of novel Barium Titanate Zinc Ferrite nanocomposite synthesized using low temperature techniques	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ / Vol 26 Issue 03, Jan 2022: 598-605	View article
23	Studying Structural and Optical Properties of Novel Aluminium Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite.	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ / Vol 26 Issue 03, Jan 2022: 606-612	View article
24	Qualitative Analysis of Magnetic Moment in BaTiO ₃ -Ni _x Fe _{3-x} O ₄ Nanocomposite Synthesized by Novel low Temperature technique	International Journal of Recent Technology and Engineering (IJRTE)/ International	1.2/ Publon (Web of Science) & UGC	Published/ Vol 10 Issue 05, Jan 2022 : 17-22	View article
25	Study of Bandgap Energy of Novel Cu _{0.5} Ba _{0.5} Fe ₁₂ O ₁₉) _{1-x} /Co _{0.6} Zn _{0.4} Fe ₂ O ₄) _x Nanocomposite	International Journal of Recent Technology and Engineering (IJRTE)/ International	1.2/ Publon (Web of Science) & UGC	Published/ Vol 10 Issue 04, Nov 2021 : 171-176	View article
26	Novel Heterostructure BTO-ZFO nanocomposite: A Quantitative Study	Rasayan Journal of Chemistry/ National	0.33/ Scopus & Web of Science	Published/ / Vol 15 Issue 04, Dec 2022: 2197-2202	View article
27	Ferrites: A study on structure, properties, synthesis, characterization, and Applications	Journal of Xidian University/ International	5.4/ Scopus & UGC Care	Published/ / Vol 17 Issue 07, Aug 2023: 1474 – 1526	View article

Research Patents

S.No.	Patent Title	Name of the Applicant(s)	Filed/Granted Patent No.	Award Date
1	Insitu Engineering mixed metal oxides on functionalized carbon cloth: A Paradigm for Flexible Electrode Material in CO ₂ Electroreduction into Formic Acid.	<u>Dr. Allwin Sudhakaran</u> , Dr. Arvind H. Jadhav, Hemavathi M, Dr. Akshaya K. Samal, Mr. Rohan Mugadum	JBR.1583	Applied on 12-08-2024

PUBLICATIONS IN CONFERENCE

S.No.	Title	Journal Name/ (National/ International)	Impact factor/ Indexing	Status/ Vol, Year, Pag.No.	Document link
1	Solar energy-based LED-powered light and inverter circuit with different types of sensors.	IOP Conference Series: Material Science and Engineering/ International	0.5/ Scopus	Published/ / Vol 1126, Mar 2021	View Link
2	Ferrites a study on structure, Properties, Synthesis and Characterization and Application	ISBN book Chapter in Conference Proceedings / International	—	Published/ / OP-206, Jul 2021: 301	View Link
3	A Review of Factors affecting Structural, Morphological, and Dielectric Properties of Barium Titanate Using Different Preparation Techniques	ISBN book Chapter in Conference Proceedings / International	—	Published/ / OP-209, Jul 2021: 304	View Link

INSTRUMENTS HANDLED

- | | | |
|----------------------------------|---|---|
| ✓ CH instrument | ✓ Scanning Probe Microscopy (SPM: AFM and MFM Mode) | ✓ Differential Scanning Calorimetry (DSC) |
| ✓ Rotary evaporator | ✓ Thermal Vacuum Evaporation coating unit (Thin Film coating) | ✓ Thermo Gravimetric Analysis (TGA). |
| ✓ Zeta potential | ✓ Fourier Transform Infrared Spectroscopy (FTIR) | ✓ Electrospinning. |
| ✓ Dynamic light scattering (DLS) | ✓ Ultraviolet-visible Spectroscopy (UV-Vis) | ✓ Photoluminescence (PL)
✓ Sophisticated Single-needle Electrospinning |

TECHNICAL & SCIENTIFIC PROFICIENCY

➤ Computational & Material Science Software

- ✓ **Quantum Chemistry & DFT:** Quantum ESPRESSO, Gaussian, Wien2k, Materials Studio, COMSOL Multiphysics, OOMMF, Advanced Nanolabo, BURAI.
- ✓ **Crystallography & Structure Refinement:** Full-Prof, MAUD, GSAS-II, JANA 2006, HighScore Plus, Match-4, Profex, TOPAS, PCPDFWIN.
- ✓ **Molecular Visualization & Modeling:** VESTA, Avogadro, ChemDraw Ultra, ORTEP, WinGX, VINUS.
- ✓ **Characterization & Spectroscopy:** OMNIC (FTIR), IRPal.

➤ Programming & Automation

- ✓ **Languages:** Python (Scientific Automation), C, C++.
- ✓ **AI & Agents:** CrewAI (Multi-agent orchestration), OpenClaw, Vibe Coding / AI-Assisted Dev.
- ✓ **Tools:** CLI-Tools, VS Code, Power Automate Desktop.

➤ Data Analysis & Visualization

- ✓ **Analysis:** OriginLab, MATLAB, QtiPlot.
- ✓ **Graphic Design:** Adobe Photoshop, CorelDRAW, FlipaClip.

➤ Computing Environments & Infrastructure

- ✓ **Operating Systems:** Linux (Ubuntu/Distros), Windows (7/10/11).
- ✓ **Infrastructure:** High-Performance Computing (HPC) Clusters, Shell Scripting.

➤ **Documentation & Research Tools**

- ✓ **Publishing:** LaTeX, Microsoft Office Suite (Word, Excel, PPT), LibreOffice.
- ✓ **Integrity:** Turnitin (Research Publication Standards).

PRESENTATIONS IN SEMINAR, CONFERENCE & SYMPOSIUM

S.No.	Title	Place	Period	Remarks
1	Unlocking Solar Hydrogen: Advancements and Challenges in PEC Water Splitting	Department of Chemistry, in association with the Centre for International Affairs (CFIA), Thiagarajar College of Engineering (TCE), Madurai, TN, India	31 July, 2026	Participated in this one-day International Research Seminar
2	Additive Manufacturing & Advanced Nanomaterials	Online- EduCare Taiwan	24 July, 2026	Participated in this One-day International webinar
3	Materials Insights: From Structural Refinement to Quantum Modelling	Thiagarajar College of Engineering (TCE), Madurai, TN, India	25-29 May, 2026	Invited as a Speaker and delivered a 1-day lecture.
4	Second National level Research Conclave 2026	Thiagarajar College of Engineering, Madurai, Tamil Nadu, India	26-27 March, 2026	Invited as Speaker
5	International Conference on Materials for Electronics, Energy, Quantum and Biomedical Technologies (MQUBIT-2026)	Thiagarajar College of Engineering (TCE), Madurai, TN, India	12-13 March, 2026	Invited as Speaker
6	Importance of Outcome-Based Education (OBE) in India and Student-Centered Learning: Annual Day Function	Government High School, V. Ramasamipuram, Vittalpatti, Sedapatti, Madurai, Tamil Nadu, India	04 March, 2026	Invited as a Chief guest and delivered an awareness lecture during the school's Annual Day Function.

7	New Trends in advance functional materials	Online- SRM Institute of Science and Technology, Ramapuram, TN, India	31 July, 2025	Participated in this One day International symposium
8	Theoretical and experimental investigation of multiferroic BaTiO ₃ - ZnFe ₂ O ₄ nanocomposite: Tailoring properties via green synthesis rote	Jain University, sponsored by SERB & DRDO	18-20 April 2024	Presented (oral) at National Conference on Sustainable Energy Conversion & storage: Functional Materials, systems & Applications (NCSECS-2024)
9	Insights into copper precursor influence on CuS catalyst synthesis for enhanced electrochemical CO ₂ reduction into formic acid	CNMS Jain University sponsored by Rasayan Sanstha, India	10 October, 2023	Presented (oral) in International Symposium on Science beyond Boundary: Invention, Discovery, Innovation, and Society (Rasayan 17)
10	Novel Heterostructure BTO-ZFO nanocomposite: A Quantitative Study	Centre for Material Science, Karpagam Academy of Higher Education	Jun 2022	Presented in International Conference on Recent Innovations in Multidisciplinary Scientific Research (ICRIMSR-2022)
11	Analyzing magnetic properties of novel CBH- NZF nanocomposite	Centre for Material Science, Karpagam Academy of Higher Education	Jun 2022	Presented in International Conference on Recent Innovations in Multidisciplinary Scientific Research (ICRIMSR-2022)
12	Quantitative Analysis of Low-temperature Novel Heterostructure BaTiO ₃ - ZnFe ₂ O ₄ Nanocomposite	Centre for Material Science, Karpagam Academy of Higher Education	Mar 2022	Presented in 1st International Conference on minerals, materials and manufacturing methods (ECM-2022)
13	Evidence of Magnetic Interaction between CuBaFe ₁₂ O ₁₉ and CoZnFe ₂ O ₄ in the nanocomposite	Department of Physics, Faculty of Arts, Science, Commerce and Management of Karpagam Academy of Higher Education	May 2022	Presented at National Conference on Advanced Materials and Its Applications (NCAMA-2022)

14	Charge Density Analysis, Structural, Electrical, and Magnetic Studies of BaTiO ₃ +NiFe ₂ O ₄ Heterostructure Composite	Department of Physics, Faculty of Arts, Science, Commerce and Management of Karpagam Academy of Higher Education	May 2022	Presented at the National Conference on Advanced Materials and Its Applications (NCAMA-2022)
15	Deliberating Structural, Optical, and Spectroscopic Properties of Novel Barium Titanate Zinc Ferrite Nanocomposite Synthesized Using Low-Temperature Techniques	Online - Indian Academicians and Researchers Association	Feb 2022	Presented at the 2nd International conference on “Science, Technology, and innovation
16	Review on Magnetic and Dielectric Properties of BaTiO ₃	SSN Research Centre-SSN Institutions	Jul 2021	Presented in an International Symposium on “Modeling of Crystal Growth Processes and Devices (MCGPD)
17	Investigating Optical Properties of Novel Heterostructure Aluminium doped Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) nanocomposite.	Online - Indian Academicians and Researchers Association	Feb 2022	Presented at the 2nd International conference on “Science, Technology, and innovation
18	Ferrites: A study on Structure, Properties, Synthesis, Characterization and Applications	Online- Indian Association for Crystal Growth (IA- CG), held at Karpagam University	Jul 2021	Presented in an International Symposium on “Modeling of Crystal Growth Processes and Devices (MCGPD)
19	Structural, Morphological, Magnetic and Microwave absorption properties in Nickel Ferrite	Online- International Organization for Crystal Growth (IOCG), held at Karpagam University	Jul 2021	Presented in an International Symposium on “Modeling of Crystal Growth Processes and Devices (MCGPD)
20	A Review of Factors affecting Structural, Morphological, and Dielectric Properties of Barium Titanate Using Different Preparation Techniques	Ramco Institute of Technology, Virudhunagar	May 2021	Presented in National level Web Conference on “Challenges and Innovations in Engineering and Technology (NWCCIET)

21	Study on structure, Properties, Synthesis and Characterization and Application of ferrite materials	Ramco Institute of Technology, Virudhunagar	Feb 2021	Presented in National level Web Conference on “Challenges and Innovations in Engineering and Technology (NWCCIET)
22	Synthesis, Characterization, properties and applications of ZnFe ₂ O ₄ Nanoparticles a Review	Karpagam University, Coimbatore.	May 2021	Presented at National Conference on “Emerging Trends in Engineering Sciences (NCETES)
23	Synthesis, Characterization, Properties and Applications of ZnFe ₂ O ₄ Nanoparticles: A Review	Online- Indian Association for Crystal Growth (IACG)	Apr 2021	Presented at International Virtual Conference on “Energy Conversion and Storage (ICECS-2021)
24	Recent Advancements in Hexagonal and Spinel Ferrites	Online- Indian Association for Crystal Growth (IACG)	Apr 2021	Presented at International Virtual Conference on “Energy Conversion and Storage (ICECS-2021)
25	New Materials for a New Age – Multiferroics.	Karpagam Academy of Higher Education, Coimbatore.	Mar 5-6, 2020	Presented this topic in National Level seminar
26	Impact of Nanotechnology on Engineering Applications	Karpagam University, Coimbatore.	Oct 15, 2019	Participated in a National level seminar
27	Advanced Materials and Applications	Karpagam Academy of Higher Education, Coimbatore.	Jan 31st – Feb 1st, 2019	Presented this topic in National Level seminar.
28	Nanoscale study of Nanocomposite BaTiO ₃ -x ZnFe ₂ O ₄	Indra Gandhi Center for Atomic Research, IGCAR, Kalpakkam	28 May – 06 July, 2018	Attended National level Summer Training Internship Program in Physics and Chemistry (STIPAC) and carried out project work.
29	Solid Oxide Fuel Cells (SOFC)	Karpagam University, Coimbatore.	20 th Feb 2018	Presented this topic in National Level seminar

30	Advanced Materials for EMI shielding	Karpagam College of Engineering, Coimbatore.	Feb 19-21, 2017	Participated in National level seminar
31	Theoretical Study on a brief history of Gravitational waves and LIGO as an Observatory	Karpagam University, Coimbatore	Feb, 2017	Presented this topic in National Level seminar.
32	Applications related to Advance Materials	Karpagam Academy of Higher Education, Coimbatore	Jan 31st – Feb 1st, 2016	Presented this topic in National Level seminar.

WORKSHOPS

S.No.	Title	Place	Period	Remarks
1	Fundamentals and Practice of Rietveld Refinement using X-ray Diffraction data	Thiagarajar College of Engineering (TCE), Madurai, TN, India	25-26 August, 2025	Invited as a Chief guest and contributed hands-on training sessions during this one-week workshop.
2	Smart application development using IoT devices (FDP)	University of Engineering and Management (UEM), Jaipur, Kolkata, India	08 July, 2025	Attended IEM-UEM Online Training and Learning Platform / UEM, India
3	Urban Photovoltaic Technology	Thiagarajar College of Engineering (TCE), Madurai, Tn, India in association with WRI & SEPA, India	25-26 June, 2025	Attended Offline workshop/ Sustainability and Energy Practitioner Association (SEPA) an Indo-German International Collaboration
4	Smart Materials Sensor and Energy Devices (SMSSED-2020)	Karpagam University, Coimbatore.	May 25-30, 2020	Attended six days International Workshop
5	Science Academies Lecture Workshop on Quantum Mechanics	PSG Krishnammal College for Women, Coimbatore	Oct 4th – 6th, 2017	Participated in a 3-day national-level workshop sponsored by IAS Bangalore, INSA New Delhi, NASI Allahabad

6	Solar Thermal and Photovoltaic Systems	Karpagam Academy of Higher Education, Coimbatore.	Sep 2017	Participated in a two-day National Workshop
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FACULTY DEVELOPMENT PROGRAMME (FDP)

S.No.	Title	Institution	Period	FDP type/ Sponsor
1	Advanced Functional Materials and emerging Technologies for Sustainable Energy, Electronics, Healthcare, Environmental Remediation, and Industrial Applications.	Thiagarajar College of Engineering (TCE), Madurai, TN, India	18-24 Nov, 2026	About to deliver speech on this One -week International FDP (Invited as Speaker)
2	Magnetic Materials for renewable energy and storage systems	Thiagarajar College of Engineering (TCE), Madurai, TN, India	22-27 Oct, 2026	Invited as a speaker for this One-week International FDP
3	Novel Functional Materials and their role in emerging technologies	Thiagarajar College of Engineering (TCE), Madurai, TN, India	19-25 Aug, 2026	Invited as a speaker for this One-week International FDP
4	Advanced Functional Materials for Energy, Environment and sustainability	Online- SONA College of Technology, Salem, TN, India	27-31 July, 2026	Participated in this One-week National FDP.
5	Emerging Technologies for Clean Energy and Water Security	Online- Ethiraj College for Women, Chennai, TN, India	27-31 July, 2026	Participated in this One-week Interdisciplinary Virtual FDP.
6	Emerging trends in Biomaterials, Materials design, and simulation techniques	Thiagarajar College of Engineering (TCE), Madurai, TN, India	19 February 2026	Invited as a speaker in a 5-day FDP and delivered a 1-day speech on “Recent Advances and Trends in Material Designs”.

7	Advances in Computational and Experimental Research in Physics	SRM Institute of Science and Technology, Ramapuram, TN, India	21 July to 01 August, 2025	Attended a two-week International FDP
8	Design, synthesis and characterization of novel materials	Indian Institute of Information Technology (IIIT), Tiruchirappalli	25-29 October, 2021	Attended AICTE Training and Learning (ATAL) / ATAL (FDP), India
9	High-Speed Optical Links and Photonic Devices for Next Generation Data Transmission	IIT Kota	Oct 11-15, 2021	Presented and participated in a national-level 5-day workshop sponsored by ATAL.
10	Emerging research advances & trends in intelligent knowledge engineering systems	Sri Sai Ram Institute of Technology (Autonomous), Chennai	30 Sep to 06 Oct 2021	Participated in a six-day international-level FDP/ Indian Society for Technical Education (ISTE)
11	Future of examinations: Bloons taxonomy and multiple assessment tools	GTN Arts College (Autonomous), Dindigul	21-23 July, 2020	Participated in a three-day National Level FDP- Faculty Enrichment Program

ONLINE COURSES & CERTIFICATION

S.No.	Course Title	Learning Platform	Instructor	Period
1	Instrumentation Program on Transition Electron Microscopy (TEM) and Fourier Transform Infrared Spectroscopy (FTIR)	Google Meet Link	Mrs. Nandhini Muthukumaran, Founder/ CEO, Nanonest Laboratory Pvt.Ltd, Coimbatore, TN, India	20-21 Nov, 2025
2	Finite Element Analysis Convergence and Mesh Independence	Coursera	Jousef Murad, Mechanical Engineer, Coursera Instructor	20 June, 2025
3	Environmental Science	IEM-UEM Teaching Learning	Dr. Ruchira Mukherjee, University of Engineering and Management (UEM), Jaipur, India.	18 June, 2025
4	Material Science and Engineering	IEM-UEM Teaching Learning	Dr. G.S. Taki, Department of ECE, Institute of Engineering and Management (IEM), Kolkata, India.	17 June, 2025

WEBINAR

International: ✓ Attended: 30
 ✓ Presented 3

National: ✓ Attended: 37
 ✓ Presented 10

BOOK PUBLICATIONS

S. No.	Title	Area/ Year	Publisher	ISBN No.
1	Basics of Major Synthesis and Characterization of Nanomaterials	Nanomaterials/ 2021	KDP Publishers	ISBN-13 : 979-8723281752
2	Fundamentals of Zeolites & Composites : Nanoscience and Nanotechnology	Nanoscience and Nanotechnology/ 2021	KDP Publishers	ISBN-13 : 979-8746228833
3	Chemistry behind Nanoscience & Nanotechnology	Nanochemistry/ 2021	KDP Publishers	ISBN-13: 979-8806630927
4	Basic Fundamentals of Graphene & Graphene sheets based Nanocomposites	Nanotechnology/ 2021	KDP Publishers	ASIN: B094T2NTV5
5	A Review on Mesoporous materials & their Nanocomposites	Nanoscience and Nanocomposites/ 2021	KDP Publishers	ASIN: B094SQSX1C
6	Briefing various Modeling techniques of polymer/ CNT Nanocomposites	Nanocomposite/ 2021	KDP Publishers	ASIN: B094R1RPDZ
7	Introduction to Cellulose, Hemicellulose & Lignin based Nanocomposites	Nanoscience and Nanocomposites/ 2021	KDP Publishers	ISBN-13: 979-8500770981

8	An Overview of CNT & their Nanocomposites	Nanoscience and Nanotechnology/ 2021	KDP Publishers	ASIN: B094S92NY5
9	Briefing Metal-Organic Frameworks (MOF) & its Composites	Material Science/ 2021	KDP Publishers	ASIN: B094T6B2G2

SKILLS

- ✓ Analytical skills: Analyzed experimental data to investigate the effects of ferroelectric and magnetic phase coupling in multi-phase ferroic composites using advanced spectroscopic and microscopy techniques.
- ✓ Critical thinking: Developed a new theoretical model to explain the coupling of ferromagnetic properties in multi-phase ferroic composites, providing new insights into the design of materials with enhanced multifunctional properties.
- ✓ Research skills: Conducted experiments to synthesize and characterize multi-phase ferroic composites with tailored compositions and microstructures to optimize their multifunctional properties for various applications.
- ✓ Technical skills: Proficient in using various laboratory equipment, including Thermal Vapour Evaporation Unit, UV, PL, FTIR spectroscopy, and advanced microscopy and spectroscopic instruments like SPM-AFM configuration. Experienced in MAUD, FULLPROF, and Origin for data analysis and modeling.
- ✓ Communication skills: Published research articles in high-impact scientific journals and presented research findings at international physics, materials science, and engineering conferences.
- ✓ Collaboration and teamwork: Worked with researchers from diverse backgrounds, including physicists, materials scientists, and engineers, on interdisciplinary research projects related to multi-phase ferroic composites in IGCAR, Kalpakkam.
- ✓ Time management and organization: Managed multiple UG and PG research projects simultaneously, ensuring that experimental timelines were met and research results were presented promptly.
- ✓ Attention to detail: Conducted careful characterization of the microstructure and properties of multi-phase ferroic composites, ensuring high-quality research outcomes and accurate reporting of research results.
- ✓ Adaptability and flexibility: I adapted research protocols to new experimental conditions and troubleshoot as needed, ensuring successful outcomes in developing and characterizing multi-phase ferroic composites.
- ✓ Intellectual curiosity: Developed research questions and pursued innovative research directions in physics and materials science, leading to publications related to multi-phase ferroic composites.

MEMBERSHIP IN PROFESSIONAL SOCIETIES

S. No.	Name of the Society	Details of Membership	Remarks	Nature of Tech. Activities done through the society
1	International Association of Engineers (IAENG)	Lifetime Member	Member ID: 303443	Published two articles in IJRTE journal suggested by IAENG
2	National Institute For Technical Training & Skill Development (NITTSD)	Lifetime Member	Member ID: NITTSD/PROF ESSOR/58855	—
3	Institute For Engineering Research and Publication (IFERP)	Member	Member ID: PM47361890	Attended webinars suggested by IFERP
4	All India Council For Technical Skill Development (AICTSD)	Lifetime Member	Member ID: AICTSD/PROF ESSOR/72288	Attended webinars suggested by AICTSD
5	National Digital Library of India (NDLI) club	Executive student member	Executive member on behalf of the students of Karpagam Arts & Science College	Suggested ideas to Improve Library facilities and infrastructure

CO-CURRICULAR ACTIVITIES

- ✓ Participated in Entrepreneurship Awareness Camp (EAC) (2016) organized by DST-NIMAT and Department of Commerce, Karpagam University, Coimbatore, TN, India.

LANGUAGES KNOWN

- English
- Tamil
- Hindi

EXTRACURRICULAR ACTIVITIES

- ✓ Part of Guinness World Record participant creating a world record in the Criteria, Largest one day Art competition (2010).
- ✓ Participated in the Guinness Book of World Records and Limca Book of World Records in “World’s Longest Drawing” (2009).

WORK IN PROGRESS

- ✓ Writing journals related to Electrocatalysis, mainly focusing on HER, OER, and ORR systems.
- ✓ Writing journals related to Electrocatalysis, mainly focusing on CO₂ utilization, reduction, fixation, and conversion into value-added products.
- ✓ Writing journals related to Magnetic and optical properties of ferrites, metal oxides, composites, and thin films
- ✓ Writing research articles related to Topological Edge states in CNT in collaboration with Madurai Kamaraj University, Madurai, Tamil Nadu, India.
- ✓ Writing articles related to Core/shell BTO-ZFO in collaboration with the Indira Gandhi Centre for Atomic Research, Kalpakkam, Tamilnadu, India.
- ✓ Writing articles related to Si-doped magnetic materials in collaboration with Vel-Tech University, Chennai
- ✓ Writing articles related to transition doped materials for advanced antibacterial properties in collaboration with Al-NahrainUniversity and Mustansiriyah University, Al-NahrainUniversity, Baghdad, Iraq.
- ✓ Writing articles related to the influence of nanoparticles at different plasma jet treatment times and their optical properties with UV-Vis-NIR in collaboration with Minia University, El-Minia, Egypt.
- ✓ Writing articles related to the biological activity of MgO nanoparticle synthesis by plasma-assisted reduction method in collaboration with Assiut University, Assiut, Egypt
- ✓ Writing articles related to silver nanoparticles before and after radiation of alpha particles for skin cancer applications in collaboration with University of Baghdad, and Mustansiriyah University, Iraq.

REFERENCES

1. **Dr. A. Saranya,**
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HOBBIES

- Drawing & Painting
- Reading
- Writing
- Meditation

DECLARATION

I, ALLWIN SUDHAKARAN, do at this moment confirm that the information given above is Accurate to the best of my knowledge.

Date:

Place:

(Dr. ALLWIN SUDHAKARAN)